

LearnPath: An Evidence-Driven Adaptive Learning System

PHASE 1 REPORT

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***“This project addresses the following Sustainable Development Goal(s):
Goal-No : 4 : Quality education.”***

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ABSTRACT

Online courses commonly present the same sequence of lessons to learners with different prior knowledge, goals and rates of progress. LearnPath is an adaptive-learning prototype that addresses this limitation by maintaining a learner model at the level of reusable skills, diagnosing knowledge from multiple pieces of evidence, checking prerequisite relationships, and organizing eligible learning opportunities into a personalized, explainable course path. Its central design principle is to distinguish what a learner has demonstrated from what the learner reports, what remains unknown, and how certain the system is about its estimate. The system combines a React and TypeScript web application, an Express API, PostgreSQL persistence and a Python intelligence-service boundary. A curriculum graph represents skills and prerequisites. A diagnostic engine selects skill-linked questions over bounded sessions and stores versioned response evidence, and a centralized mastery service updates mastery, confidence, evidence state and retention estimates. The application then evaluates prerequisite readiness, generates candidate skills and creates an explainable path. Graph analytics and a Random Forest benefit model are kept separate from authoritative learner-state and prerequisite decisions, so a statistical score can never override a required prerequisite. The project is evaluated through software verification and synthetic engineering experiments. These are useful for comparing policies and checking invariants but are not evidence of diagnostic accuracy or improved learning among real students. The results reveal an important limitation: the adaptive diagnostic policy can stop before adequate coverage, while a fixed-length policy achieved higher synthetic coverage. LearnPath is therefore presented as an implemented research prototype whose calibration, educational effectiveness and fairness require prospective evaluation with consenting learners.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
BKT	Bayesian Knowledge Tracing
CAT	Computerized Adaptive Testing
DAG	Directed Acyclic Graph
FK / PK	Foreign Key / Primary Key
JSON	JavaScript Object Notation
KLI	Knowledge-Learning-Instruction
ML	Machine Learning
NDCG	Normalized Discounted Cumulative Gain
RF	Random Forest
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 PROJECT OVERVIEW

Digital learning platforms make a broad range of courses accessible, but access alone does not ensure that a learner is working on the right material. A beginner may need foundations, while an experienced learner may need only a short verification before a more advanced topic. A uniform sequence can spend time on content already understood while missing a prerequisite gap that makes later material difficult.

LearnPath is an adaptive-learning system that represents a course as a collection of skills and relationships rather than only as a list of lessons. It gathers evidence from diagnostic questions, practice, quizzes, assessments and retention checks, and maintains one global knowledge estimate for each learner–skill pair. When the same skill appears in more than one enrolled course, evidence about that skill is reused across those courses; enrollment tracks course-specific progress and never creates a competing copy of the learner’s knowledge.

The system uses this learner state to support a chain of decisions. It identifies which skills have evidence, which need further assessment and which are not yet assessed; checks required prerequisites before admitting a skill to the set of learnable candidates; ranks the eligible opportunities; and produces a course-specific path with a clear next action, upcoming skills, and locked skills accompanied by reasons. After the learner answers questions or completes practice, new evidence is recorded and the path can be regenerated from the updated state.

LearnPath also aims to make personalization *visible*: the interface presents knowledge state, confidence, coverage, prerequisites, recommendations and reasons, so a learner can see which evidence and constraints shaped each recommendation.

1.2 PROBLEM STATEMENT

In a conventional course, content is organized in a predetermined order shared by all learners. This has several limitations. It may not account for prior knowledge gained through other courses, work or independent study. A short quiz provides weak evidence about a large curriculum when it samples only a few skills. A correct answer does not necessarily establish durable mastery, and an incorrect answer may reflect an isolated mistake rather than a stable gap. Finally, later skills depend on

earlier ones, so recommending a topic merely because it appears relevant or has a high model score can produce an invalid learning sequence.

The problem addressed by this project is therefore:

“To design and develop an adaptive learning system that combines incomplete and differently reliable evidence about a learner’s knowledge with a curriculum’s prerequisite structure to recommend a defensible, explainable next learning action — without treating self-report as proof, missing evidence as zero knowledge, or a recommendation score as a substitute for an unmet prerequisite.”

1.3 OBJECTIVES

The primary objective of LearnPath is to build an evidence-driven personalization pipeline whose decisions can be inspected. The major objectives are:

1. **Global skill-level learner model** — represent knowledge per learner–skill pair so that demonstrated knowledge is reused across courses.
2. **Progressive diagnostic evidence** — gather evidence in bounded sessions, preserve the distinction between assessed, estimated and unknown knowledge, and identify when more questions are needed.
3. **Prerequisite graph** — represent curriculum relationships as a directed graph and use required edges to decide whether a skill is ready to learn.
4. **Explainable personalized path** — generate a dependency-aware path from eligible skills and current evidence, with a reason for every recommendation and every lock.
5. **Transparent learner interface** — communicate the recommendation, its rationale, confidence, progress and outstanding prerequisites.

Supporting objectives are an auditable history of question versions and responses, retention estimated separately from mastery, shared skills across enrollments, and evaluation that never presents simulated outcomes as real student results.

1.4 SCOPE OF THE PROJECT

The scope of LearnPath covers the following functional areas:

- Learner registration, sign-in, course discovery and enrollment, with an optional self-portrait of prior familiarity.
- Course and skill curricula, diagnostic programs, question selection and server-side scoring.

- Global skill state, an append-only evidence ledger, and retention estimates.
- Prerequisite analysis, candidate generation, ranking and explainable path generation.
- Learner-facing progress views, and a Python service boundary for model inference and graph analytics.

LearnPath is a research and demonstration prototype. It is not a validated high-stakes examination system, an accredited certification platform, or evidence that a particular intervention improves educational outcomes. The recommendation model was evaluated on synthetic interactions, and the diagnostic stress tests use simulated response patterns. These results support engineering comparisons only; accuracy against a reference test, learning gains and fairness across real learner groups require studies with consenting learners.

1.5 PROPOSED APPROACH

LearnPath models a learner's knowledge as a global state over curriculum skills. Each state records an estimated mastery value, a confidence value, an evidence classification and a supporting evidence history. Self-reported familiarity can guide the order in which skills are checked, but it is never converted directly into mastery. Performance observations are stored with their source and question context, and a centralized update service uses them to revise the learner state.

The curriculum is a set of skills with directed prerequisite relationships. Before recommendation, the application evaluates required relationships against the learner's state; skills with unmet required prerequisites remain locked, while advisory relationships only influence ordering. Eligible skills are ranked and a path planner presents a dependency-valid sequence with explanation data. The diagnostic engine reports what was sampled and never converts an untested skill into a low score.

CHAPTER 2

LITERATURE SURVEY

2.1 ADAPTIVE LEARNING SYSTEMS

Adaptive learning systems use information about the learner, the instructional content and learning activity to adjust what is presented or recommended. The Knowledge-Learning-Instruction framework describes how domain knowledge, a learner model, instructional events and learning outcomes interact [1]. This is relevant to LearnPath because recommendations should not be derived from course structure alone: the system also needs an estimate of the learner's current knowledge and an explicit account of how that estimate was obtained.

Knowledge tracing models how a learner's knowledge changes with practice. Bayesian Knowledge Tracing estimates the probability that a skill is mastered while allowing for guesses, slips and learning transitions [2]. LearnPath shares this skill-level perspective but uses a centralized evidence ledger and a policy-based update that combines several evidence sources; it is not described as Bayesian Knowledge Tracing because that algorithm is not what the implementation executes. Computerized adaptive testing selects items from prior responses to gain information from few questions; the global-information approach of Chang and Ying is one example [3]. LearnPath also selects questions progressively, but its aim is course-skill diagnosis with coverage, application evidence, contradiction resolution and bounded assessment programs. It does not claim to be a calibrated item-response-theory test.

2.2 LEARNER DIAGNOSIS AND KNOWLEDGE ASSESSMENT

Assessing knowledge from few responses is inherently uncertain. One item shows only that a question was answered correctly or incorrectly; it does not establish stable mastery of a broad skill. Items that vary in difficulty, cognitive demand and context provide stronger evidence, which motivates LearnPath's repeated and varied observations, explicit confidence, and classifications such as *probed*, *needs confirmation* and *not tested*. Self-reported familiarity is separated from demonstrated performance: it may prioritize a verification or foundation probe, but subsequent performance evidence decides the skill model.

Diagnostic quality also depends on correct skill mapping, clear answer keys and unambiguous wording; item statistics are descriptive monitoring until a representative sample and external validation exist. Retrieval practice research

shows that testing supports later retention [6], and distributed-practice research motivates revisiting knowledge over time rather than treating one successful session as permanent mastery [7]. LearnPath reflects this architecturally by tracking later evidence and a retention estimate, although the present experiments do not validate a particular spacing schedule or forgetting curve.

2.3 KNOWLEDGE GRAPHS AND LEARNING RECOMMENDATIONS

Prerequisite relationships form a directed graph in which skills are nodes and dependencies are edges. Such edges give a planner a structural constraint, and research has explored inferring prerequisite relations from course structures [4]. LearnPath stores curriculum relationships explicitly and evaluates them in application logic. Graph structure supports more than ordering: topological layers expose curriculum stages, descendant reach indicates how many later skills depend on a foundation, and centrality highlights gateways. NetworkX provides such algorithms in Python [5]; in LearnPath they are derived and read-only, enriching explanations and breaking close ranking decisions without deciding whether a prerequisite is satisfied.

Personalized learning-path research studies how to sequence learning objects to match learner needs, for example by fusing knowledge-graph embeddings, sequence mining and collaborative filtering [10]. LearnPath combines the general idea with an explicit separation among evidence estimation, prerequisite eligibility, candidate ranking and path presentation, so it is clear why a skill is available, why another is locked, and which component contributed to the ordering. The ranking model is a Random Forest [8] and its ordering quality is reported with Precision@k and NDCG@k [9].

2.4 EXISTING APPROACHES AND THEIR LIMITATIONS

Table 2.1 summarizes the main existing approaches and the limitation of each with respect to the problem stated in Section 1.2.

Approach	Strength	Limitation for this problem
Fixed course sequence	Simple, predictable	Ignores prior knowledge and prerequisite gaps
Short placement quiz	Quick entry point	Samples few skills; weak evidence for a large curriculum
Knowledge tracing [2]	Models learning over time per skill	Needs fitted parameters; does not by itself enforce curriculum prerequisites
Computerized adaptive testing [3]	Efficient item selection	Targets ability estimation; needs calibrated item banks
Graph / path recommenders [4], [10]	Capture dependencies and sequencing	A model score can override a hard prerequisite unless gated

Table 2.1 Existing Approaches and Limitations

2.5 RESEARCH GAP

The reviewed research establishes foundations for learner modeling, adaptive item selection, prerequisite representation and recommendation. A practical system must still integrate these without confusing their roles: a single score should not simultaneously stand for knowledge, confidence, prerequisite readiness and expected benefit; a model rank should not override a hard dependency; and an absent response should not be misrepresented as failure. LearnPath addresses this *engineering* gap by organizing the workflow into distinct stages — global learner-state estimation, diagnostic evidence collection, prerequisite validation, eligible-candidate generation, benefit ranking and path generation — and preserving evidence provenance so each decision can be inspected. This is an integration and implementation contribution, not a claim that adaptive learning or knowledge graphs are new.

CHAPTER 3

PROPOSED SYSTEM

3.1 SYSTEM OVERVIEW

LearnPath provides an end-to-end workflow that turns course enrollment and learner evidence into a revisable personalized path. A learner selects a course and may provide a self-portrait of prior familiarity. The system schedules a diagnostic program, presents one question at a time, and stores responses against versioned questions and mapped skills. A mastery service updates the global learner–skill state; the prerequisite service then determines which skills are eligible, which remain locked and what evidence is missing; candidate generation and ranking order the eligible opportunities; and a path service persists an explainable sequence for the enrollment. New evidence can make the path stale, after which it is regenerated.

The system separates *course progress* from *knowledge*. Completing a lesson records activity, but completion alone does not prove mastery. A learner may know a skill through another course, so global knowledge can be recognized in several course contexts while each course keeps its own enrollment and module status.

3.2 SYSTEM REQUIREMENTS

3.2.1 Functional Requirements

ID	Functional Requirement	Description
FR01	Registration and Sign-in	Learners register and authenticate before accessing the system.
FR02	Course Enrollment	Learners browse courses and enroll; progress is tracked per enrollment.
FR03	Self-Portrait	Optional familiarity statements are stored separately from performance evidence.
FR04	Diagnostic Sessions	Bounded assessment sessions are scheduled over a course backlog of skills.
FR05	Question Selection and Scoring	Versioned skill-linked questions are selected; answers are validated and scored on the server.
FR06	Evidence and Mastery Update	Each response is appended to an evidence ledger and applied to the global learner–skill state.
FR07	Prerequisite Evaluation	Required edges lock downstream skills; recommended edges only influence ordering.
FR08	Candidate Ranking	Eligible skills are ranked using model benefit, revision and graph signals.
FR09	Path Generation	An explainable, enrollment-specific path is persisted and can be regenerated.
FR10	Learner Views	Dashboard, Skill Passport, prerequisite view, practice and retention views.

Table 3.1 Functional Requirements

3.2.2 Non-Functional Requirements

ID	Non-Functional Requirement	Description
NFR01	Integrity	Mastery, eligibility and ranking are computed on the server; the browser is not an authority.
NFR02	Explainability	Every recommendation and lock carries a stored reason and decision-time evidence.
NFR03	Auditability	Evidence is append-only; question versions are snapshotted with each attempt.
NFR04	Security	Correct answers are hidden before submission; options are validated against the question.
NFR05	Reliability	If model validation fails, the service reports unavailability instead of fabricating an ML result.
NFR06	Modularity	Web client, policy API, persistence and intelligence service evolve through explicit contracts.
NFR07	Scalability	Graph traversal is linear, $O(V+E)$; ranking is $O(K \log K)$ over K eligible skills.
NFR08	Data Separation	Synthetic research data never enters the application learner database.

Table 3.2 Non-Functional Requirements

3.3 SYSTEM ARCHITECTURE

The web client is implemented with React and Vite and presents course selection, diagnostic assessment, results, the Skill Passport, prerequisite views, practice, retention and personalized paths. It requests state and decisions from backend services. The Express application API, written in TypeScript, coordinates authentication, enrollment, diagnostics, evidence, mastery updates, prerequisite checks, candidate generation and path persistence. PostgreSQL is the authoritative store. A separate Python FastAPI service provides bounded intelligence capabilities — Random Forest benefit inference and read-only NetworkX graph analytics — behind a schema- and version-validated boundary. The Python service does not own learner state and cannot bypass the required-prerequisite check.



Figure 3.1: System Architecture

Diagnostic selection, prerequisite enforcement and path orchestration are modules inside the TypeScript API, not additional deployed services. The intelligence service receives only graph projections or feature vectors for candidates that have already passed prerequisite eligibility; its outputs return to the API for validation, and the model never writes learner state directly.

3.4 LEARNER AND CURRICULUM MODELS

The learner model uses a global learner–skill identity. Each record holds **mastery** (the estimate of demonstrated knowledge), **confidence** (how much evidence supports the estimate), an **evidence state** (unknown, estimated, assessed or verified) and retention information. *Unknown* is a meaningful state: a skill that was not sampled is never assumed to be zero. Performance evidence is stored in an append-only ledger recording the source, skill, response or score, difficulty, attempt context and before-and-after state, so that a result can be explained later. Question content used in an attempt is versioned, so later edits to the question bank cannot silently change the meaning of an earlier response.

The curriculum model contains courses, modules, skills, course–skill mappings and directed prerequisite relationships. A *required* prerequisite can block a downstream skill; a *recommended* one only influences ordering. Course enrollment owns course-specific progress, while the global skill state is reused whenever the same skill appears in another course.

3.5 DIAGNOSTIC AND RECOMMENDATION DESIGN

The diagnostic program manages assessment across a course that may contain more skills than one sitting can test. It keeps a durable backlog and schedules bounded sessions: in comprehensive mode a session focuses on at most four skills and the backlog is revisited until skills are sufficiently assessed or the question bank is reported as blocked, while quick placement samples across course regions without claiming complete coverage. Question selection considers coverage needs, skill-level evidence, item metadata and unresolved contradictions, and the learner can see the coverage boundary.

After evidence is recorded, the prerequisite service evaluates required edges against global mastery and evidence. Unknown or weakly supported knowledge never silently satisfies a dependency. Candidate generation then separates locked, eligible and already-covered skills *before* the statistical ranker runs. The ranking layer estimates relative learning benefit for eligible candidates only, using a Random Forest [8] with a fixed, versioned feature contract that was trained and tested on simulated learner interactions split by learner identity. Precision@k and NDCG@k [9] describe ordering quality against synthetic labels; they do not give the probability that a real learner will benefit. The path service combines the ranking result with policy and graph context, stores a reason, and keeps prerequisite eligibility authoritative.

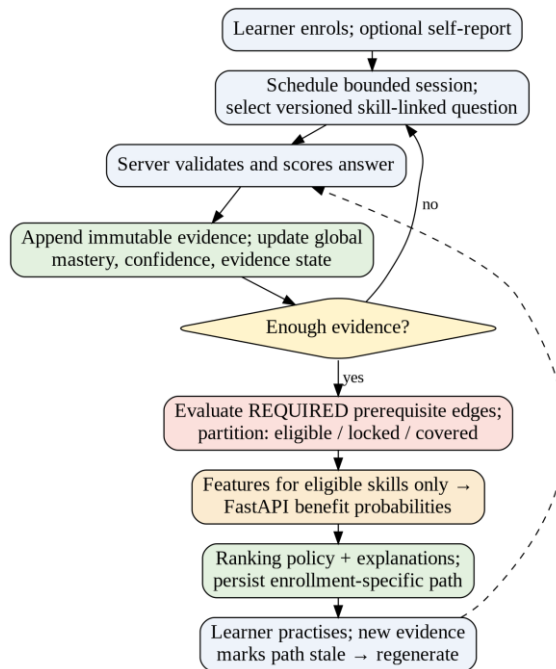


Figure 3.2: Evidence-to-Path Data Flow

Self-report is non-authoritative: it can influence probe order and a small, confidence-damped cold-start ranking prior, but never creates mastery. The offline research pipeline is deliberately separate from this runtime flow: a deterministic simulator reads a curriculum snapshot, a frozen 55-feature contract is applied, learners are split into disjoint train, validation and test groups, models are trained and compared, and a checksummed artifact is produced without inserting simulated learners into the application database.

3.6 DATABASE AND TECHNOLOGY STACK

PostgreSQL stores users, courses, modules, skills, course–skill mappings, prerequisite edges, enrollments, diagnostic programs and attempts, versioned questions, answers, global learner–skill mastery, immutable evidence, practice and retention state, and personalized path versions. A uniqueness rule on learner and skill prevents the same knowledge from being duplicated per course, and foreign keys with server-side validation ensure answers refer to the question’s actual options. Neo4j is not required: authoritative curriculum relationships live in PostgreSQL and NetworkX performs derived analysis.

Layer	Technology	Responsibility
Web client	React, Vite, TypeScript	Learner workflows and visualization of decisions
Application API	Express (TypeScript)	Orchestration, mastery updates, prerequisite gate, path planning
Persistence	PostgreSQL	Authoritative learner, curriculum, evidence and path records
Intelligence service	Python, FastAPI	Benefit inference and read-only graph analytics
Graph analytics	NetworkX	Layers, descendant reach, centrality, bottlenecks
Ranking model	Random Forest	Benefit probability for eligible candidates

Table 3.3 Technology Stack

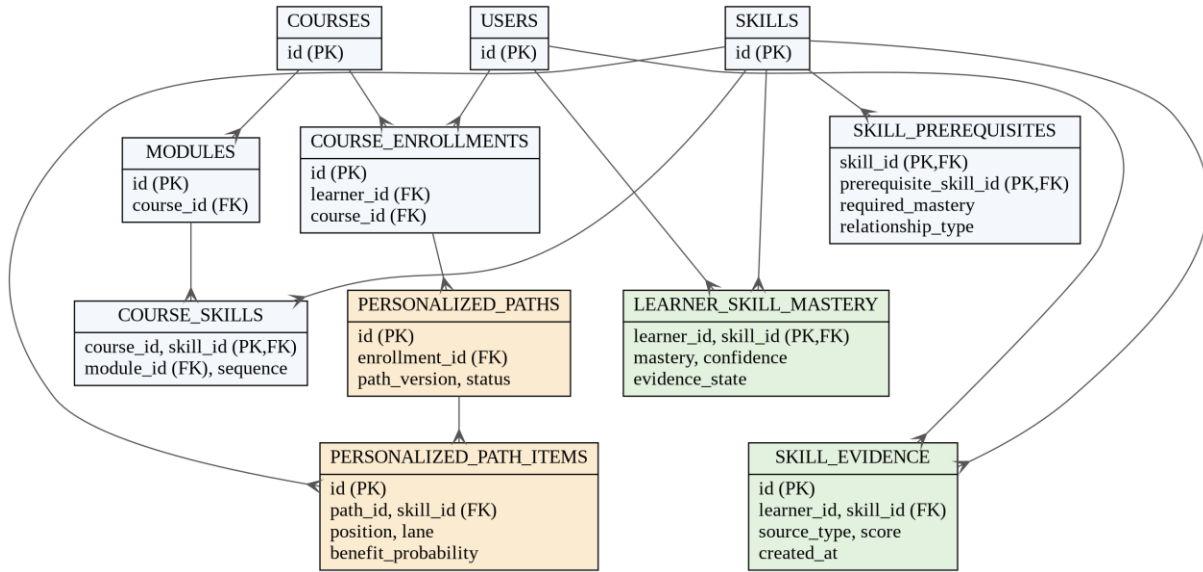


Figure 3.3: Conceptual Relational Data Model

The central cross-course invariant is the unique learner–skill mastery record. Enrollment and path versions are course-specific, while skill evidence and the current estimate are reusable. The prerequisite table is a directed self-relationship over skills; the edge type and required-mastery threshold determine whether it blocks progression.

CHAPTER 4

MODULE IMPLEMENTATION

4.1 CURRICULUM AND LEARNER MODEL

Course content is represented through reusable skills. A course maps its modules and resources to skills, and prerequisite edges relate the skills. Because the learner's state is keyed by learner and canonical skill — not by enrollment — a skill demonstrated in one course can inform readiness in another that references it. The model stores more than a mastery number: confidence and evidence state let the interface and planner distinguish a stable estimate from a tentative one. A course-completion event updates progress but never mastery, so attendance or opening a resource is not confused with demonstrated learning.

State updates are centralized. Diagnostic answers, practice, assessments and retention checks all pass through one mastery service: evidence is appended with its source and context, then the unique learner–skill row is updated. This supports longitudinal review and avoids incompatible scoring logic in individual modules.

4.2 DIAGNOSTIC ASSESSMENT

The diagnostic starts from the course scope and assessment intent. Self-portrait statements are stored apart from performance evidence; they may bring a skill forward or suggest a challenge item, but cannot mark it as mastered. For large courses, assessment is an *assessment program*: it tracks which regions have been sampled, keeps a backlog of unresolved skills, and tells the learner when a skill has not been assessed rather than inventing a low value.

Questions are linked to one or more skills and carry difficulty, cognitive level and an item role. *Anchor* items give an initial signal, *verification* items resolve contradictions and *challenge* items test a stronger claim, so the engine collects more than a count of right and wrong answers. Answers are validated on the server against the options of the selected question, the correct answer is not exposed before submission, and the submitted question version is preserved so the result can be reproduced. The learner-facing classification is deliberately stricter than a raw percentage (Table 4.1).

Status	Rule (summary)
NOT_TESTED / PROBED	No measurements / exactly one measurement.
NEEDS_CONFIRMATION	Mixed evidence, or an unresolved critical competency.
READY	≥ 3 consistent correct responses, application evidence, ≥ 2 cognitive levels, ≥ 2 competencies, performance ≥ 0.80 or course target.
MASTERED	≥ 3 measurements, a correct application/analysis response, ≥ 2 cognitive levels, mastery $\geq$ course target and 0.80, confidence ≥ 0.70 , competency coverage ≥ 0.80 .
GAP / FORGOTTEN	Low mastery with repeated negative observations and no correct response / previously strong skill with at-risk retention and poor current performance.

Table 4.1 Diagnostic Classification Rules

4.3 KNOWLEDGE GRAPH AND PREREQUISITES

The graph holds skill nodes and directed *required* or *recommended* edges. The prerequisite service loads the course scope and supporting ancestors and checks each required edge against the learner’s global state. A required edge participates in eligibility; a recommended edge informs sequencing or explanation only. When a prerequisite is not sufficiently demonstrated, the downstream skill is returned as locked together with the unmet skill and an evidence reason; when another course has already satisfied it, the dependency is recognized without repeating the assessment.

NetworkX supplies derived analysis — topological layers, downstream reach, centrality, prerequisite routes, bottlenecks and counterfactual unlocks. These are read-only and never satisfy a prerequisite. For eligible skills the graph service combines three normalized signals into a bounded gateway score, where D is the descendant ratio, B the normalized betweenness centrality and U the immediate counterfactual-unlock ratio:

$$G = 0.50 \cdot D + 0.30 \cdot B + 0.20 \cdot U$$

The score contributes at most $0.06 \cdot G$ to path priority after required-edge validation. If NetworkX analytics are unavailable, the TypeScript service still enforces required prerequisites and the graph bonus is omitted.

4.4 CANDIDATE RANKING AND PERSONALIZED PATHS

Candidate generation uses the latest global state for the enrollment and includes course skills and, where necessary, supporting prerequisite skills outside the direct module mapping. Candidates are partitioned into eligible, locked and covered, because a ranking algorithm must never make an invalid skill learnable by assigning it a high score. Eligible candidates may receive a benefit estimate from the Random Forest inference interface, which validates feature names, ordering, bounds and version. The interface presents the score as a ranking signal, not a guarantee of learning gain.

The path planner converts ranked candidates into a dependency-aware sequence. Each path item stores the skill, course context, status, reason codes, decision-time mastery and confidence, prerequisite information and model provenance. Visual states are *recognized/completed*, *current*, *recommended next*, *upcoming* and *locked*; locked items show which prerequisite needs attention. Paths are tied to an enrollment: new evidence marks the path stale, the learner regenerates it, and earlier versions are preserved for comparison. Multiple course paths stay distinct while shared-skill recommendations are coordinated through the global skill identity.

4.5 LEARNING EVIDENCE AND RETENTION

The mastery service applies bounded updates using source and performance context, records the before-and-after state, and updates confidence and evidence classification from the accumulated observations. The source weights (Table 4.2) are explicit engineering policy values, not empirical claims about the relative validity of each source.

Evidence source	Diagnostic / assessment / module assessment	Quiz	Practice	Retention check
Weight	0.40	0.30	0.25	0.35

Table 4.2 Source Weights in the Mastery Update

Retention is represented separately from mastery: mastery is what has been demonstrated, retention estimates how accessible it still is after time has passed. LearnPath uses an exponential decay form in which $R(t)$ is the retention after elapsed time t , M_0 the mastery at the latest relevant evidence and λ_{eff} an effective decay parameter reduced by confidence and evidence volume:

$$R(t) = M_0 \cdot e^{(-\lambda_{eff} \cdot t)}$$

Revision is recommended only when mastery was previously demonstrated; a learner is never described as having forgotten a skill that was never shown. Retention checks reuse assessment questions and record evidence through the same mastery service, whereas opening a lesson does not reset the evidence timestamp. The decay rate is an engineering estimate and has not been shown to predict real forgetting.

4.6 USER INTERFACE AND DATA MANAGEMENT

The interface is organized around decisions and feedback. Course selection leads into a diagnostic or the current path; the dashboard summarizes active courses, the next action, progress and global skill state. The path view is the central visualization, showing recognized, current, recommended, upcoming and locked skills with prerequisite explanations. The diagnostic presents one question at a time and its results view explains what was assessed, where evidence is strong or mixed, and which skills still need checking. The Skill Passport makes cross-course recognition visible, and practice and retention interactions return before-and-after state. All displayed values are sourced from backend responses — the frontend never invents mastery, confidence, path order or benefit values — and loading, empty, unavailable and error states explain what is happening.

4.7 CORE ALGORITHMS AND PSEUDOCODE

The pseudocode below summarizes the implemented decision sequence in implementation-neutral form; it does not replace the versioned source. Thresholds are configurable prototype parameters, not calibrated educational constants.

Algorithm 1. Progressive session planning and item selection

The backlog is ordered by coverage priority (unassessed, needs refresh, needs confirmation), session count, self-report verification priority and curriculum position; self-report affects ordering only. A quick placement spreads at most four skills across up to four module regions, whereas a comprehensive session takes up

to four skills from the highest-priority module, with a question limit of min(28, available questions). Within a session the item selector works in stages:

```
stage <- COVERAGE    IF sequence <= MIN(8, skill_count, budget)
  VERIFICATION IF contradictory evidence OR sequence >= MIN(22, budget)
  CONFIRMATION otherwise
score(q) <- weighted sum of bounded signals (info gain, uncertainty, coverage,
  gateway impact, diversity, misconception, retention risk, relevance,
  self-report) - exposure penalty - topic-switch penalty
RETURN highest score, ties by curriculum order then stable question id
SHOULD_CONTINUE: false at budget/bank exhaustion; true while responses < MIN(12,
pool),
  observed skills < MAX(8, CEIL(0.75*skills)), a high-impact contradiction remains,
  or a top-three prerequisite gateway is unresolved
```

The item-value weights in the current policy version are information gain 0.25, competency uncertainty 0.16, decision impact 0.15, gateway importance 0.15, uncertainty 0.15, course relevance 0.10, retention risk 0.08, evidence diversity 0.07, misconception value 0.05 and self-report priority 0.05, with a further 0.18 for verification items when evidence is mixed. These are policy parameters, not learned or universally optimal values. A response with item difficulty d (bounded 1–5) receives performance $0.72 + 0.05d$ if correct, $0.12 + 0.035d$ if “unsure” and $0.03 + 0.05d$ if incorrect; diagnostic reliability grows with independent measurements, cognitive-level diversity, application evidence and consistency, and is capped at 0.78.

Algorithm 2. Evidence-based mastery and confidence update

```
PROCEDURE UPDATE_LEARNER_SKILL(previous, performance, source, reliability):
  performance, reliability <- CLAMP(., 0, 1)
  IF previous.mastery is UNKNOWN: start <- 0.5 ; w <- 1.0 * reliability
  ELSE: start <- previous.mastery ; w <- SOURCE_WEIGHT[source] * reliability
  mastery <- CLAMP(start * (1 - w) + performance * w, 0, 1)
  confidence <- CLAMP(0.08 + 0.45*(1 - EXP(-n/6)) + 0.05*MIN(diversity,3)
    + 0.12*MIN(sessions/4,1) + 0.06*MIN(avg_difficulty/5,1)
    + 0.12*consistency + 0.10*[retention evidence], 0, 1)
  IF all evidence is DIAGNOSTIC: confidence <- MIN(confidence, 0.45)
  evidence state <- UNKNOWN | ESTIMATED | ASSESSED | VERIFIED (by thresholds)
  APPEND immutable evidence (before/after) ; UPDATE unique learner-skill row
  MARK affected enrollment paths stale ; RETURN mastery, confidence, state
```

For an unknown skill the first update equals the observed performance when reliability is 1.0; lower reliability scales it. Later observations move the prior mastery toward the new performance by the source weight of Table 4.2.

Algorithm 3. Confidence-aware prerequisite gate and candidate partition

```

PROCEDURE EVALUATE_AND_PARTITION(skills, edges, learner_state):
  ASSERT required-prerequisite graph is acyclic
  FOR each skill, FOR each REQUIRED edge into skill:
    s <- learner_state[edge.prerequisite]
    effective <- UNKNOWN if s.mastery unknown; s.mastery if confidence unknown;
      else MAX(0, s.mastery - (1 - s.confidence) * 0.10)
    IF effective is UNKNOWN OR effective < edge.required_mastery: add to missing
  IF missing not empty: LOCKED (with named prerequisite reasons)
  ELSE IF mastery meets course target (or recent placement ready) and no revision due:
    RECOGNIZED (global or provisional)
  ELSE: ELIGIBLE
  gateway <- 0.50*D + 0.30*B + 0.20*U if analytics valid, else 0
  RETURN eligible, locked, covered, dependency-valid order

```

Algorithm 4. ML-assisted personalized-path construction

```

PROCEDURE GENERATE_PATH(enrollment):
  eligible, locked, covered <- partition from Algorithm 3
  predictions <- FastAPI.predict(versioned features of ELIGIBLE skills only)
  VALIDATE exact candidate coverage, feature version and response schema
  FOR each eligible c:
    priority[c] <- MIN(1, predictions[c] + 0.08*[revision due]
      + 0.06*gateway[c] + need(familiarity)*(1 - confidence))
  SORT by priority, probability, dependency level, course sequence, name
  path <- recognized skills + current skill + RECOMMENDED NEXT (first ranked)
    + UPCOMING (rest) + LOCKED (missing prerequisites, null model score)
  PERSIST enrollment-scoped path version with snapshots, provenance, reasons

```

The reported-need prior is 0.18 for NEVER_LEARNED, 0.12 for KNOW_A_LITTLE, 0.06 for UNSURE, 0.04 for COMFORTABLE and 0.02 for VERY_COMFORTABLE, multiplied by (1 – confidence), so familiarity never acts as mastery and loses influence as evidence accumulates.

Algorithm 5. Retention projection and Algorithm 6. Offline model training

```
PROCEDURE PROJECT_RETENTION(mastery, confidence, n, last_evidence, now):
  IF mastery UNKNOWN OR n = 0: RETURN UNKNOWN, revision_due = false
  lambda <- base_lambda * (1.15 - 0.5*confidence) / (1 + MIN(0.12*LOG(1+n), 0.45))
  R <- CLAMP(mastery * EXP(-lambda * days_since(last_evidence)), 0, 1)
  state <- STRONG (R>=0.75) | MODERATE (R>=0.55) | AT_RISK (R>=0.30) | CRITICAL
  revision_due <- mastery >= 0.65 AND state in {AT_RISK, CRITICAL}

PROCEDURE TRAIN_AND_EVALUATE(curriculum_snapshot, seed=42):
  rows <- simulate_learner_candidate_interactions ; transform with frozen 55 features
  split BY learner identity: 70% train / 15% validation / 15% test
  fit Random Forest, Gradient Boosting, Logistic Regression on train
  select model maximizing validation NDCG@5 (ties: ROC-AUC, F1)
  evaluate selected model ONCE on test; compare with Highest-Skill-Gap and Popularity
  persist model, checksums, split membership, versions and metrics (synthetic label)
```

Training and evaluation never update the PostgreSQL learner model; any future training on real learner data would need a separate consent, governance and validation protocol. For adjacency lists, one DAG traversal with dependency levels costs $O(V+E)$ and sorting K eligible candidates costs $O(K \log K)$, excluding feature construction, database access and inference. These are algorithmic bounds, not measured latency.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 SYSTEM DEMONSTRATION AND USER INTERFACE

The application is demonstrated as a continuous learner workflow. A new learner creates an account, browses and enrolls in a course, and optionally states prior familiarity. The learner then starts the course diagnostic, answers server-selected questions, and receives a result that separates demonstrated evidence from uncertain or unassessed areas. The system updates the global Skill Passport, checks the prerequisite graph, generates a course-specific path and names a recommended next skill. After studying or practising, the learner submits more evidence and regenerates the path to see whether the recommendation or the locked states change.

The demonstration exposes the intended decision chain: what has been demonstrated, what remains uncertain, which prerequisites are met, what is eligible to learn, and why one eligible skill is recommended. It is a functional software demonstration, not a controlled study of student learning.

LearnPath

LEARNING THAT REMEMBERS YOU

One account. A path shaped around you.

- ✓ Your skill knowledge follows you across courses
- ✓ Start each course at the right lesson
- ✓ Practice turns progress into clear next steps

Choose a course · Find your starting point · Keep growing

← Back to LearnPath

CREATE YOUR ACCOUNT

Start learning at the right place

Choose a course, show what you already know, and get a learning path built around you.

Display name

Email

Password

Confirm password

✓ Your account and learning progress are kept secure.

Create account →

Already registered? Sign in

Figure 5.1: Login screen

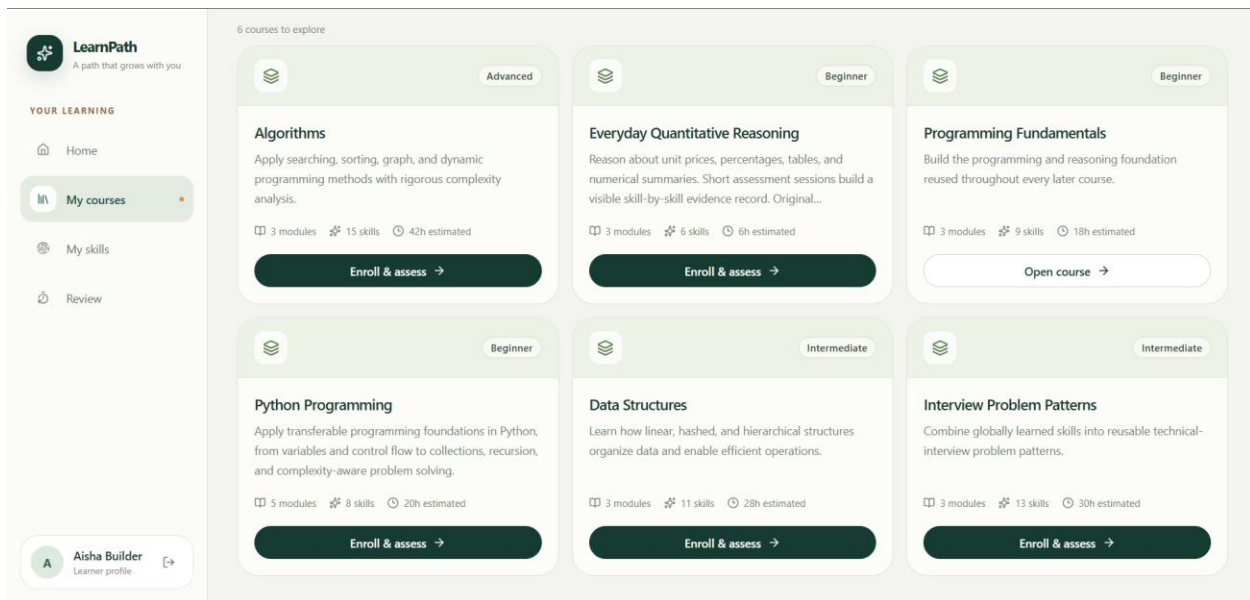


Figure 5.2: Learners Dashboard

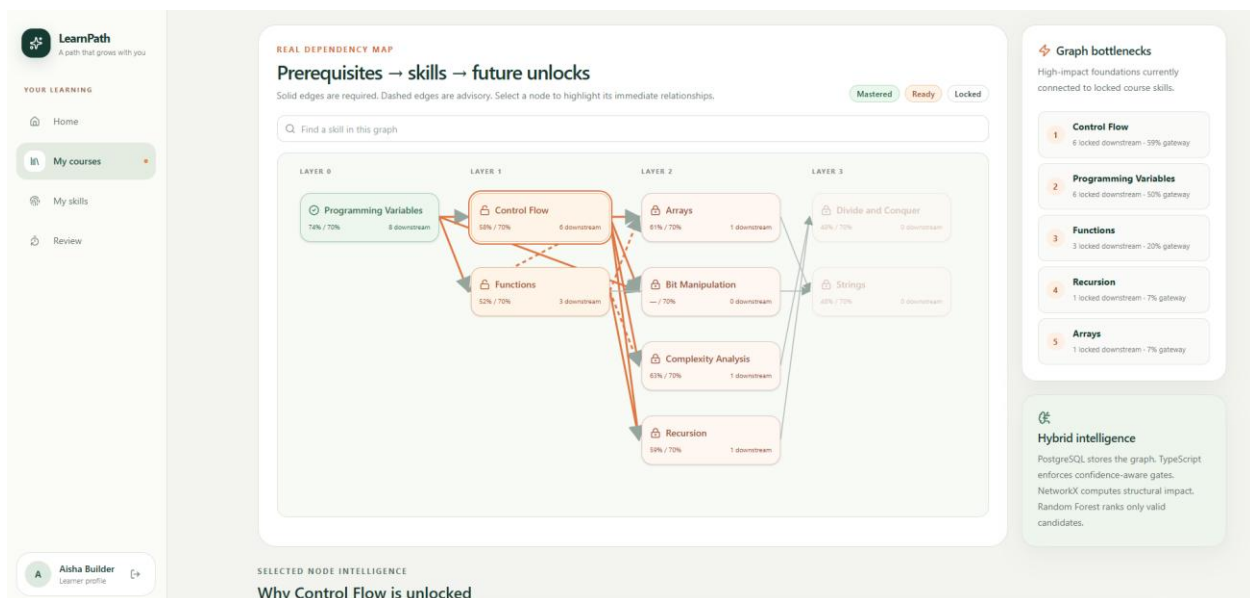


Figure 5.3: Knowledge graph

5.2 IMPLEMENTATION VERIFICATION

Verification covers individual services and end-to-end invariants: a learner–skill pair stays globally unique across enrollments; answers are rejected when an option does not belong to the question; the correct answer stays hidden until submission; diagnostic evidence updates mastery only through the centralized service; unknown skills remain unassessed rather than zero; required prerequisites block downstream

candidates; a high recommendation score can never bypass a lock; the same skill is recognized across courses; and a path can be regenerated after new evidence. Diagnostic quality monitoring and engineering audits report test status, item sample size, policy invariants and artifact provenance. Such verification shows that the software follows its encoded rules for the tested cases; it does not show that the question bank measures every skill accurately, that mastery scores are calibrated for students, or that recommendations improve learning.

5.3 EXPERIMENTAL RESULTS

5.3.1 Diagnostic Policy Experiment

The diagnostic experiment is a synthetic engineering stress test using four-skill sessions, question-bank depths of three and six, four response-propensity profiles, two noise conditions and 256 random seeds per configuration. This produced 4,096 paired synthetic response matrices and 24,576 policy runs of at most 28 questions each. The compared policies were the current adaptive policy, an eligibility-first policy, fixed-length policies, a matched-question comparison, a no-self-report condition and a no-graph condition; 2,000 stratified bootstrap replicates were used across the 16 experiment cells. Because the responses were generated, the intervals describe variation under the simulation design and are not confidence intervals for real students.

Policy	Mean questions	Mean coverage	Remark
Adaptive (current)	10.07	0.2655	No selectable question (premature stop) in about 47.02% of runs
Eligibility-first	12	0.3381	No premature selector-null events
Fixed-12	12	0.3777	Highest coverage; exceeds eligibility-first by about 0.0396
Matched-question comparison	—	0.3046	Question count matched to the comparison policy
No self-report	—	unchanged	No observed coverage change in this simulation
No graph input	—	0.2790	Slight increase over adaptive, not the expected decrease

Table 5.1 Diagnostic Policy Results (Synthetic)

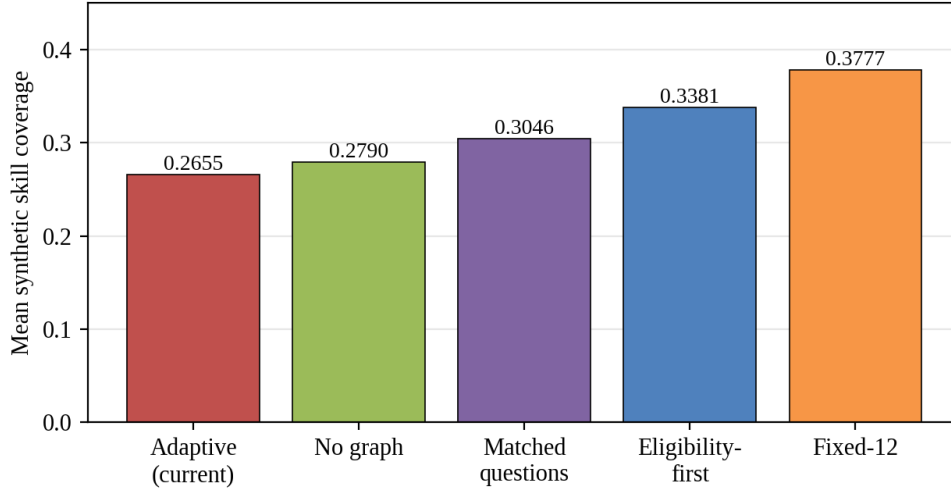


Figure 5.4: Mean Synthetic Skill Coverage by Policy

Eligibility-first coverage exceeded the adaptive policy by about 0.0726 (stratified bootstrap interval of approximately [0.0687, 0.0765]) at an average cost of about 1.93 additional questions. These outputs act as a diagnostic-policy warning: the current stopping and selection interaction needs investigation. They do not show which policy is more accurate, because coverage is measured against the generated response model, not against independently observed knowledge. The same package reported 136 valid invariant combinations with no recorded violations, and planner traversals were exercised at curriculum sizes of 100, 1,000 and 5,000 skills (25, 250 and 1,250 sessions under the assumption that each scope resolves). These support bounded software-path and scheduling checks only; they are not a benchmark of educational quality, production load or response time.

5.3.2 Offline Recommendation-Model Evaluation

The offline evaluation used synthetic interactions from 1,000 simulated learners and 20,000 candidate interactions, with learners separated into train, validation and test groups. On held-out learners the Random Forest achieved ROC-AUC 0.801, Precision@5 0.525 and NDCG@5 0.578, slightly below the Highest Skill Gap baseline at 0.581 (Figure 5.2). The result is mixed: the model separates synthetic benefit labels to a degree, but its ranking metric did not surpass the strongest baseline. Because outcomes and labels are synthetic, the numbers cannot be read as real-world learner benefit, and any claim that a running path used a deployed model must be checked against the runtime artifact and deployment record.

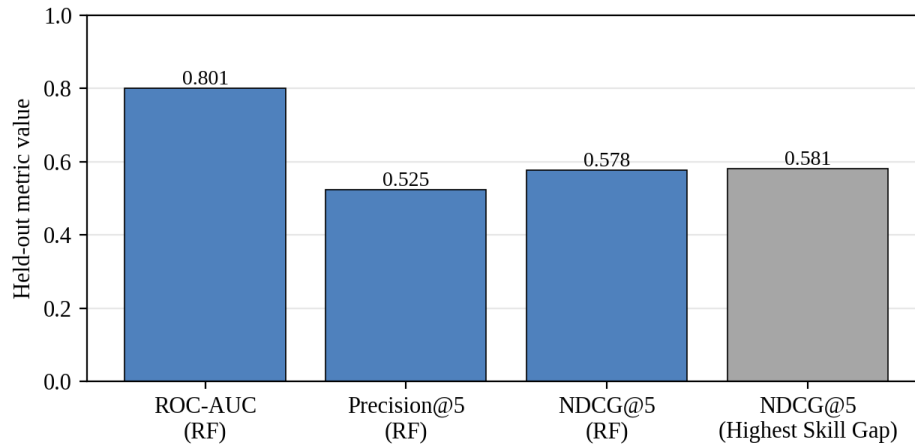


Figure 5.5: Held-out Random Forest Metrics (Synthetic Data)

5.4 DISCUSSION AND LIMITATIONS

The most important result is not a claim of high diagnostic accuracy but evidence that policy choices materially affect which skills are assessed. The adaptive selector sometimes stopped before sufficient coverage, whereas a longer fixed assessment reached more. A policy that saves questions is attractive, but it should not stop while important course regions remain unresolved. The next engineering step is to diagnose why the selector reaches a null candidate, strengthen backlog and coverage guarantees, and re-evaluate under the same frozen protocol. The experiment also cautions against assuming that self-report or graph features automatically improve diagnosis: self-report did not change coverage, and removing graph input slightly increased it. These results may reflect the designed response model or implementation interactions and call for targeted ablation, not broad conclusions.

Several limitations remain. Synthetic profiles cannot represent the diversity, misconceptions, language differences, motivation or test-taking behavior of real students. The question bank's content validity and skill mapping need expert review. Mastery and retention policies are configurable engineering models rather than calibrated psychometric ones, and the recommendation labels come from a simulated benefit formula. There is no prospective study against an independent reference assessment, no delayed retention test and no subgroup fairness analysis, and an incorrect prerequisite edge can lock a learner unnecessarily or permit an inappropriate sequence. LearnPath should therefore be described as a working adaptive-learning prototype with inspectable evidence and recommendation workflows, not as a system that perfectly knows a student's mastery or has proven superior learning outcomes.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

LearnPath implements an adaptive-learning workflow that connects learner evidence to personalized course sequencing. It maintains global skill-level knowledge, treats self-report as a prior rather than proof, supports progressive diagnostic coverage, records evidence with provenance, evaluates prerequisites before ranking, and presents a course-specific path with reasons. It distinguishes course progress from demonstrated knowledge, separates mastery from confidence and retention, and lets skills shared across courses be recognized in more than one enrollment.

The architecture separates the web interface, application policy, PostgreSQL persistence and Python intelligence boundary, which keeps the main pipeline inspectable: diagnosis updates evidence, prerequisite checks determine eligibility, ranking orders only eligible skills, and path generation explains the sequence. NetworkX provides structural analysis without becoming the authority for readiness, and the Random Forest evaluation is an offline capability on synthetic data, not validated educational benefit.

The engineering evaluation revealed areas for improvement rather than grounds for an effectiveness claim: the selector's premature stopping shows that assessment efficiency must be balanced against coverage, and the model's NDCG@5 did not exceed the strongest baseline. The project provides a concrete foundation for evidence-driven personalization while keeping implementation verification distinct from educational validation.

6.2 FUTURE WORK

6.2.1 Diagnostic Coverage and Stopping

The first priority is to guarantee that high-priority course regions are either assessed or explicitly left unresolved, to give every remaining backlog item a durable reason, and to handle question-bank exhaustion without ending the assessment as if knowledge were settled. The revised policy should be re-evaluated against fixed-length and coverage-oriented baselines under a frozen protocol.

6.2.2 Curriculum and Question Review

Subject experts should validate skill definitions, required and recommended edges, question-to-skill mappings, answer keys, cognitive levels and misconception tags. Item analysis needs sufficient response counts and should keep descriptive statistics distinct from calibrated measurement; item-response or knowledge-tracing models can be examined where data and assumptions allow.

6.2.3 Prospective Learner Study

With ethics review and informed consent, a study could compare diagnostic estimates with an independent reference assessment, run delayed retention checks, test whether explanations are understood and measure whether learners complete recommended paths, using predefined outcomes and a justified, adequately sampled fairness analysis.

6.2.4 Recommendation Model, Usability and Accessibility

The model should be evaluated on governed real outcome data with learner-disjoint partitions, strong non-ML baselines, calibration analysis and metrics aligned with the product objective, while the eligibility gate stays in place. Cold-start personalization can use learner goals and the self-portrait provided these never become unverified mastery claims. Usability and accessibility testing with real users should accompany each of these steps.

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Abstract : learning systems are only as trustworthy as the learner estimates behind them, and those estimates should show how much evidence supports them. This paper describes LearnPath, an implemented diagnostic architecture that keeps mastery, confidence, assessment coverage, and prerequisite readiness as separate quantities. Self-reports decide which questions get asked but never count as mastery evidence, and short bounded sessions feed a persistent course-wide assessment program. We also test whether preference-based question selection keeps candidate items available and improves evidence coverage, using controlled response replay: six policy configurations Adaptive on 4,096 paired synthetic response matrices, giving 24,576 session runs. The deployed selector ends sessions prematurely in 47.02% of these cases. An eligibility-first variant removes the failure without touching the scoring or confidence rules, yet its assessed-skill coverage (33.81%) still trails a stratified twelve-question baseline (37.77%). The results expose a scheduling defect and a bounded fix; they do not show adaptive superiority or diagnostic accuracy, and we make no claim about human learning outcomes. Finite invariant checks and planner traversal complement the replay

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Index Terms—Adaptive diagnosis, assessment coverage, evidence provenance, knowledge graphs, learner modeling, personalized learning.

I. INTRODUCTION

A personalized learning system has to answer two questions that are easy to run together: what should this learner study next, and what can the system honestly claim to know about them? Recommending an advanced topic is risky when its foundations lack evidence, yet repeatedly serving introductory material the learner already knows is just as unhelpful. The hard part is building a usable picture of the learner from evidence that is partial, mixed, and sometimes contradictory.

Large curricula make this harder. A fixed entrance quiz can sample only a small share of a course with many skills, and treating the unsampled skills as failures mistakes missing evidence for evidence of a gap. The opposite error is just as easy to make: propagating a correct advanced answer back through a prerequisite graph invents foundations nobody observed. Learners may recall a sophisticated procedure yet have lost a basic operation, so a diagnostic model needs room

for such cases instead of forcing every learner onto one line from beginner to expert.

LearnPath takes an evidence-bounded approach. It gives each learner a provisional starting region, keeps unresolved skills on record, and continues assessment over bounded sessions and later learning activity. Self-reported familiarity shapes which evidence the system goes looking for, but only scored responses are recorded as evidence. Skills have a shared identity so evidence can be reused across courses, while the current course decides which skills and prerequisites matter.

Estimation and item choice are well studied in knowledge tracing, adaptive testing, and constrained selection [1]–[6]. The problem we address is a narrower engineering one that surfaced in our own system: applying a preference filter before checking eligibility can empty the candidate pool. A preferred skill that has hit its observation cap can hide another skill’s usable item, and the selector then reports an exhaustion that does not exist. We isolate this ordering defect and do not present eligibility or uncertainty as new research topics.

We ask three questions. RQ1: does the selector return an item whenever at least one unasked item is still eligible? RQ2: how does assessed coverage compare with stratified selection at fixed and at matched question counts? RQ3: how do coverage and failure behavior change with bank depth, response propensity, self-report, and graph metadata? None of them tests diagnostic validity.

Our contributions are: (1) an auditable architecture that keeps mastery, confidence, coverage, provenance, retention, and prerequisite readiness apart; (2) the identification of the preference-before-eligibility failure, with a correction that preserves availability; and (3) a paired synthetic replay of deployed, corrected, stratified, and ablated configurations, reported together with the cases where the correction does not help. The correction has been evaluated offline only and has not replaced the deployed policy.

II. RELATED WORK AND POSITIONING

Knowledge tracing infers skill acquisition from observed performance [1], and Deep Knowledge Tracing applies recurrent networks to learning histories [2]. LearnPath uses neither; its update rules are explicit and not fitted to an independent cohort. Evidence-centered assessment ties the assessment argument to its evidence and operational processes [7]; LearnPath keeps those records inspectable, though traceability alone does not show that its items measure the intended constructs.

Computerized adaptive testing (CAT) offers information-based selection [3], severely constrained selection [4], and constrained full-test assembly [5]. Shadow-test methods handle exposure control through item-ineligibility constraints [6], and Han separates content balancing, the selection criterion, and exposure control [8]. LearnPath does not claim to have introduced any of this. Our contribution is the narrower property that a preference filter must not remove every usable alternative while local eligibility is nonempty. We do not solve globally feasible test assembly or formal exposure control, and our information proxy uses author-supplied priors rather than Fisher information; the exposure penalty and fixed weights are engineering rules, not a calibrated CAT model.

Knowledge-space theory describes the feasible knowledge states of a domain [9]. LearnPath instead treats authored prerequisite edges as learning constraints and assessment priorities, and a correct advanced answer never establishes the prerequisites beneath it. Ngo *et al.* recommend learning paths by combining graph embedding, sequence mining, and collaborative filtering [10]; their task differs from ours, since we gather the evidence a path planner consumes, and we do not reproduce their model as a baseline. Uncertainty-aware knowledge tracing uses stochastic representations [11], and domain-informed models can propagate mastery and uncertainty through subjective logic [12]. We claim neither predictive superiority nor novelty in uncertainty handling, and an explicit confidence formula is not calibration, which needs empirical agreement with observed outcomes [13]. Trainable spaced repetition [14] targets recall over time, whereas our cold-start replay only reads retention metadata. We evaluate item availability and internal evidence coverage, not prediction accuracy or learning effectiveness.

III. PROBLEM FORMULATION AND ARCHITECTURE

A. Curriculum and Learner State

We model the curriculum as a graph $G = (V, E)$. Each vertex is a globally identified skill, and a directed required edge from a to b means that a supports access to learning b . A course picks an ordered subset of V , groups it into modules, and sets target mastery thresholds. Several courses can reference the same skill identifier, but authors must confirm that the construct and expected level truly match before evidence is reused.

$$x_{u,s} = (m_{u,s}, c_{u,s}, z_{u,s}, a_{u,s}, r_{u,s}, H_{u,s}). \quad (1)$$

In (1), m is mastery, c confidence, z a diagnostic classification, a assessment coverage, r retention state, and H the evidence history including provenance. Prerequisite readiness is derived separately from the required edges and the learner evidence that supports them. These quantities are not interchangeable: mastery and confidence are bounded summaries rather than calibrated probabilities, a null mastery means no direct knowledge evidence, and the internal first-update prior of 0.5 does not represent measured 50% knowledge. The aim of a session is to reduce consequential uncertainty within a budget, not to maximize skills labeled mastered: find a defensible starting point, respect required prerequisites, and keep unresolved questions for later evidence.

B. End-to-End Processing

Fig. 1 shows the implemented division of responsibilities. A React front end collects goals and self-portrait responses and presents bounded assessments. A TypeScript/Express service manages program state, selection, scoring, and learner-model updates, with curriculum mappings, item metadata, attempts, snapshots, and evidence in PostgreSQL. A personalized-path service reads the resulting global skill state; the separate Python recommender in the wider application is not the diagnostic estimator studied here.

A learner picks a course and reports familiarity, and the assessment program schedules a small skill scope. Within a session the diagnostic service selects an item, records the response, recomputes priorities, and applies its stopping policy. On submission, answers are scored against the attempt snapshot, evidence is aggregated per skill, global mastery and confidence are updated, and affected paths are refreshed.

C. Self-Portrait and Evidence Provenance

In the self-portrait, learners rate familiarity from never learned to very comfortable and add confidence and experience context. Reported knowledge raises verification priority and reported unfamiliarity favors an anchor probe; neither grants mastery, removes a prerequisite, or completes coverage. Whether reported familiarity was borne out by performance can also be recorded, which explains why a learner was asked to verify a skill they called familiar.

Each active question carries a skill mapping, diagnostic role, cognitive level, difficulty, response format (single choice, multiple selection, or numeric), discrimination and guessing priors, a construct identifier, and a content version. Attempt snapshots preserve the question and scoring metadata as presented, submitted answers are immutable, and each diagnostic result keeps before-and-after learner summaries, evidence strength, and a decision explanation. Historical backfills and editable self-report records still need care when rebuilding a fully version-consistent study dataset.

IV. EVIDENCE-BOUNDED DIAGNOSTIC METHOD

A. Progressive Coverage of Large Courses

An outer assessment program keeps a persistent backlog over the whole course. A session covers at most four skills and

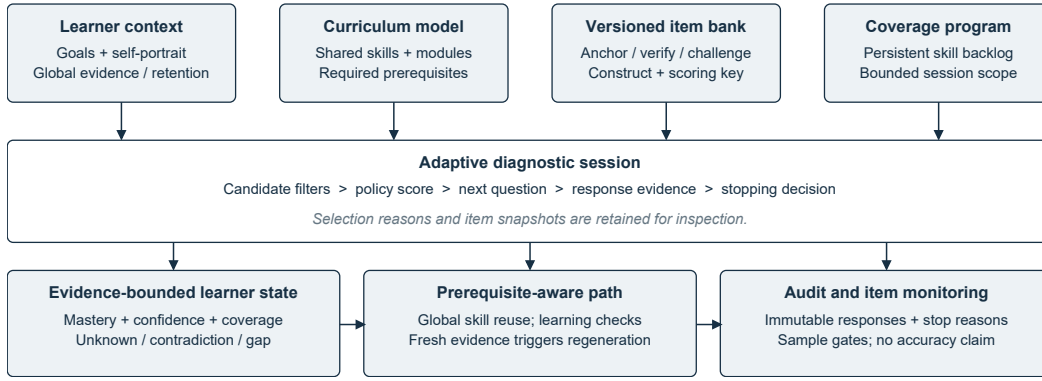


Fig. 1. LearnPath diagnostic architecture. Self-report and curriculum metadata guide collection; scored responses, not self-reports, supply diagnostic mastery evidence. Path and monitoring services consume the recorded result.

respects the diagnostic question limit. Comprehensive mode groups work by module, whereas quick placement samples curriculum regions far apart instead of stopping at the first weak foundation. Skills with unresolved evidence or retention risk stay eligible for later sessions, and since there is no fixed session limit, the per-session question budget is not a budget for the whole course. Coverage counts distinct question identifiers and evidence diversity, so repeating an item adds nothing. The usual positive route needs three distinct items, application evidence, two difficulty bands, and confidence of at least 0.50. Consistent diagnostic-only evidence has a separate route at 0.45 that does not certify mastery, and previously verified evidence has its own. A confirmed GAP or FORGOTTEN establishes coverage from two distinct items. Running out of fresh items does not by itself confer coverage. The architecture is subject-agnostic at the level of skill identifiers, but the current item formats suit quantitative reasoning best; essays and open-ended judgement are not assessed automatically.

B. Adaptive Item Selection

Within a session the policy moves through coverage, confirmation, and verification stages. Coverage avoids resampling an observed skill, confirmation seeks another independent observation, and verification favors challenge or application evidence and revisits mixed responses. Concretely, the deployed selector removes items already asked, prefers unobserved skills during coverage and contradictory skills during verification, prefers the stage-specific role within each skill (with a fallback), scores the remainder with (2), discards non-finite scores, and ranks by score, course order, and item ID, returning null for an empty pool. Curriculum role and cognitive level are separate fields, because a challenge label alone does not prove that an item measures application; the policy favors application or analysis evidence when it is missing.

$$S(q) = 100 [\mathbf{w}^T \mathbf{f}(q) + b(q) - e(q) - \tau(q)]. \quad (2)$$

The feature vector in (2) has nine components: course relevance, decision impact, evidence diversity, gateway influence, an information proxy, misconception value, retention risk,

uncertainty, and self-report priority, with weights 0.10, 0.15, 0.07, 0.15, 0.25, 0.05, 0.08, 0.15, and 0.05 in that order. The weights sum to 1.05, so S is a relative policy score and not a probability. The verification bonus b is 0.18 for mixed evidence, the exposure penalty e is capped at 0.16, and switching skills costs $\tau = 0.035$. These values predate the replay and were not fitted to its outcomes, but the repository does not record why they were chosen, so they have no empirical justification; calibration and targeted ablation remain future work.

$$I(q) = \text{clip}_{[0,1]} \left(\frac{d(q)[1 - g(q)]}{1.8} \right). \quad (3)$$

Here $d(q)$ and $g(q)$ are the item’s discrimination and guessing priors, and (3) is our information proxy. Gateway influence is the clipped dependent count divided by four. Uncertainty rises when mastery is absent, confidence is low, only one response exists, or responses disagree. Evidence diversity rewards a new cognitive level, missing application evidence, and fit to the desired difficulty. These design choices need ablation and calibration before any claim of optimal selection. A skill is limited to three observations, or four when evidence is mixed or application evidence is missing. Prior knowledge counts as recognized only if mastery reaches the course target, confidence is at least 0.70, and no at-risk or critical retention flag is present, unless a probe is explicitly requested. This conservative rule keeps a weakly supported score from suppressing assessment of a shared skill.

C. Performance and Mastery Update

For an item of difficulty $k \in \{1, \dots, 5\}$, the measurement value is $0.72 + 0.05k$ for a correct response, $0.12 + 0.035k$ for an unsure non-correct response, and $0.03 + 0.05k$ for any other incorrect one. The scoring API treats unsure as a non-correct response with no answer and rejects an answer submitted together with unsure. These values encode an ordinal

engineering judgement about difficulty and are not learned response probabilities. Each response then receives a weight

$$v_j = \text{clip}_{[0,1]}[(0.55 + 0.09k_j) \cdot (d_j/1.5) \ell_j g_j^* u_j]. \quad (4)$$

where d is clipped to $[0.25, 2.5]$, the cognitive weight ℓ is 0.62 for remember, 0.76 for understand, 0.92 for apply, and 1.00 for analyze, the guessing adjustment g^* is one minus the guessing prior for a correct response and one otherwise, and the certainty adjustment u is 0.72 for an unsure response and one otherwise. Skill performance P is the average of the measurement values weighted by $\max(0.1, v)$. Distinct item IDs are an operational independence check, not proof of statistical independence.

$$\rho = \min\{0.78, 0.16 \min(n, 3) + 0.06 \min(L, 3) + 0.08A + 0.06C\}. \quad (5)$$

Reliability ρ depends on the response count n , the number of distinct cognitive levels L , an indicator A for application or analysis evidence, and an agreement indicator C that is one when all responses are correct or all incorrect. Its ceiling of 0.78 deliberately stops a single session from being treated as conclusive, even if every answer is correct.

$$m' = (1 - \alpha) m_0 + \alpha P. \quad (6)$$

At the first diagnostic update, m_0 is the internal neutral prior and $\alpha = \rho$; when mastery already exists, $\alpha = 0.40\rho$. Updates are aggregated per skill at submission, so answer rows may share batch-level before-and-after values that are not independent per-question gains. Score improvements across sessions must not be reported as a treatment effect without a controlled study.

D. Confidence, Classification, and Uncertainty

Confidence combines evidence count, source diversity, session count, average difficulty, consistency, and the presence of retention evidence. The count term is $0.45(1 - \exp(-n/6))$; the other terms contribute up to 0.15 for source diversity, 0.12 for sessions, 0.06 for difficulty, 0.12 for consistency, and 0.10 for retention, on a base of 0.08. Consistency is one minus twice the population standard deviation of the stored per-item performance scores, clipped to $[0, 1]$, and the total is clipped to $[0, 1]$ and rounded to four decimals. If every evidence source is diagnostic, confidence cannot exceed 0.45, and with no evidence it is zero.

Fig. 2 summarizes the classification. No observation gives NOT_TESTED and one gives PROBED. Mixed correct and incorrect observations normally give NEEDS_CONFIRMATION. At least two negative observations with no correct response and mastery below 0.50 give GAP, and poor performance on a retention-sensitive check of previously strong knowledge can give FORGOTTEN. If an advanced skill scores 0.70 or higher while a required prerequisite from the same attempt is GAP or FORGOTTEN, the result is FRAGILE_FOUNDATION, so prerequisite

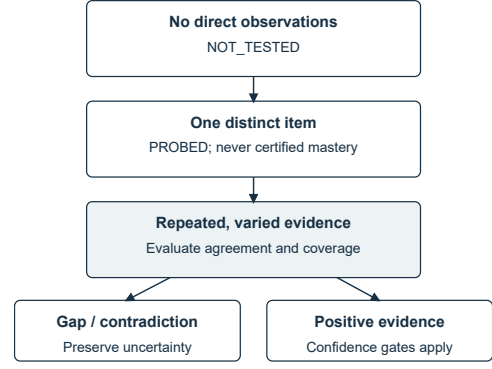


Fig. 2. Diagnostic classifications distinguish a probe from a supported decision. Coverage and evidence provenance are separate; positive evidence does not bypass confidence or prerequisite gates.

mastery is never fabricated. READY needs mastery of at least 0.65, confidence of at least 0.50, and two correct responses. MASTERED needs at least three observations, a correct application-level response, two cognitive levels, confidence of at least 0.70, and mastery meeting both 0.80 and the target. Under the current ceiling, diagnostic-only evidence therefore cannot produce a positive classification for a wholly new learner; the coverage planner has its own consistent-evidence route so assessment can progress without relabeling provisional knowledge as mastery.

$$B(m, c, n) = [\text{clip}(m - \delta), \text{clip}(m + \delta)], \quad (7)$$

$$\delta = 0.06 + \frac{0.22(1 - c)}{\sqrt{\max(1, n)}}.$$

The band in (7) is a heuristic visualization of uncertainty, not a calibrated 95% confidence interval or a Bayesian credible interval. It narrows by design as confidence or observation count grows, and its empirical coverage has not been established. Here clip bounds values to $[0, 1]$, c is clipped before use, and endpoints are rounded to four decimals.

E. Adaptive Stopping and Path Integration

A session normally needs at least 12 responses, fewer when the bank is smaller. The configured 16–22 question target is advisory and 28 is the hard maximum. Coverage requires $\min(K, \max(8, \lceil 0.75K \rceil))$ observed skills, where K is the active scope in the candidate set rather than the whole course, which means all four skills in a progressive session. Beyond coverage, the policy examines up to three highest-dependent prerequisite gateways. It continues while a gateway has an unresolved two-response disagreement, and it requires recognized evidence or at least two gateway observations (three if the evidence is mixed). A gateway with positive evidence also needs application or analysis evidence, and repeated negatives may preserve a lock. When these checks pass, the session stops as EVIDENCE_SUFFICIENT.

That label reflects an operational decision about coverage and selected gateways; it does not guarantee that non-

Algorithm 1 Eligibility-first research variant

- 1: Compute the same stage and scoring components as the deployed policy.
 - 2: Remove asked items and all items with non-finite scores.
 - 3: Apply coverage and contradiction preferences to this eligible set; restrict only when the preferred subset is nonempty.
 - 4: Apply role preference within each remaining skill, retaining its eligible fallback when the role is absent.
 - 5: Rank the remaining items with the unchanged score and tie-breaks.
 - 6: Return null only when the eligible set is empty.
-

gateway skills are resolved, and `MAXIMUM_REACHED` and `BANK_EXHAUSTED` stay distinguishable from it. After submission, the global Skill Passport shares new evidence with every enrolled course that uses the same skill. Required-prerequisite checks stay separate from diagnostic scope, so weak basics can lock advanced learning without blocking diagnostic probes of advanced skills. The knowledge-boundary summary picks the first course-ordered skill not supported by strong evidence, a conservative heuristic rather than a probabilistic frontier, and path regeneration then explains what changed the next action.

V. ELIGIBILITY-PRESERVING SELECTION

A. Preference Filtering Can Hide Eligible Questions

The deployed selector first drops questions already asked, then favors contradictory skills or a preferred diagnostic role, and only afterward removes candidates with non-finite scores caused by an observation cap. These steps do not commute. A preferred contradictory skill may already sit at its four-observation limit while another skill still has an eligible question; narrowing the pool first throws away the only usable alternatives, and the application service then logs `BANK_EXHAUSTED` although the physical bank is not empty.

The requirement is conditional availability: if at least one unasked item has a finite policy score in the current state, the selector must return an item. It need not ask every item or override the stopping policy; the rule applies only after the stopping policy has requested another question. As an illustration (not another experiment), take a verification stage in which s_1 has contradictory evidence but has reached its observation cap, while s_2 still has an unasked item with a finite score. Preference-first selection narrows to s_1 , removes its capped items, and returns null, hiding s_2 . Eligibility-first selection removes s_1 's capped items before applying preferences, so s_2 survives and is returned. The research variant changes a single candidate-pool expression so that it runs before the existing preferences; scoring weights, response processing, mastery updates, stopping rules, and confidence ceilings are unchanged. For a fixed state, let U be the set of unasked items with finite scores. Every later conditional preference keeps a nonempty subset whenever U is nonempty,

the role preference keeps at least one item per retained skill, and ranking a finite nonempty set always yields an item, so availability is preserved. This is a selection property, not a proof of optimal assessment or accurate estimation, and it says nothing about the feasibility of later selections under whole-test content or exposure constraints.

B. Scope of the Correction

The variant is an isolated source transformation applied by the research harness, not a production change, and it tests only whether eligibility ordering explains premature null selection. It does not fix shallow banks, a poorly calibrated learner model, or an inadequate stopping rule, and a session that stops as `EVIDENCE_SUFFICIENT` can still leave non-gateway skills uncertain.

VI. REPRODUCIBLE ENGINEERING EVALUATION

A. Implementation and Experimental Design

The deployed core is revision `b20a00e` with policy curriculum-evidence-adaptive-v6. A recheck on 24 September 2026 passed 182 API tests in 34 files and a read-only structural audit of 162 active diagnostic items across 42 skills; integration-test files are excluded, and these counts say nothing about learner validity. The replay runs the production pure functions for selection, stopping, estimation, confidence, and coverage without database access.

We generate 4,096 synthetic potential-response matrices from two bank depths, four response-propensity profiles, two noise settings, and 256 seeds per stratum. Each session scope has four skills, matching the production cap, and each bank holds three or six distinct items per skill. The six-item bank uses difficulty [1, 2, 3, 3, 4, 5], cognitive levels [remember, understand, apply, apply, analyze, apply], and roles [anchor, anchor, verification, verification, challenge, challenge]; the three-item bank keeps positions 1, 3, and 5. Discrimination is fixed at one, the guessing floor is 0.20, and the prerequisite dependent counts are [3, 2, 1, 0]. These banks are controlled stress-test settings, not a representative curriculum sample, and their metadata schedules exercise selector branches rather than reflect validated item calibrations.

The four profile vectors are [0.20, 0.20, 0.20, 0.20], [0.85, 0.85, 0.85, 0.85], [0.20, 0.85, 0.85, 0.85], and [0.85, 0.65, 0.35, 0.15], standing for low, high, weak-foundation, and uneven response propensity; they are not ground-truth mastery. For propensity θ and item difficulty k , the potential correctness probability is

$$p = 0.20 + (0.80 - \eta) \cdot \text{sigmoid}[\text{logit}(\theta) - 0.35(k - 3)]. \quad (8)$$

with noise η equal to 0 or 0.15. An independent unsure response with probability 0.10 suppresses correctness, mirroring the scoring contract, and nothing is learned or forgotten within a replay. Each item gets one fixed potential response per matrix that every method reuses, so this is a policy stress test, not a validated learner simulator. Skills with propensity of at least 0.60 report `COMFORTABLE` and the rest

TABLE I
SYNTHETIC POLICY COMPARISON

Policy	Questions	Assessed	Null*
Deployed adaptive	10.07	26.55%	47.02%
Eligibility-first	12.00	33.81%	0.00%
Stratified fixed-12	12.00	37.77%	0.00%
Stratified matched	10.07	30.46%	0.00%
No self-report	10.07	26.55%	47.02%
No graph components	10.08	27.90%	47.07%

*Premature null with an eligible alternative. Assessed is coverage, not accuracy. Each row uses 4,096 synthetic matrices.

NEVER_LEARNED, always with high self-report confidence. Mastery and confidence start unknown, which excludes returning learners. The protocol, seed rules, source hashes, generator, run-level CSV, aggregate JSON, and a replayable failure trace accompany the paper.

B. Baselines, Ablations, and Outcomes

Every matrix is run under six configurations. The deployed adaptive policy and the eligibility-first variant use the existing stopping rules with a maximum of 28 questions. Fixed-12 asks three stratified questions per skill in round-robin order, preferring anchor, application, and challenge items first. Matched asks as many questions as the deployed run asked on that matrix, in stratified order. No-self-report removes familiarity and confidence metadata, and no-graph sets dependent counts to zero in both selection and stopping, removing the graph-aware components together. All configurations share one estimator and coverage rule.

The main outcome is the fraction of the four skills labeled ASSESSED, which can include confirmed gaps. We also report question count, unresolved fraction, observed skills, positive READY/MASTERED declarations, and premature-null frequency, meaning the deployed selector returns no item while the eligibility-first selector finds an eligible one in the same state. A premature null is neither a false-mastery rate nor a human dropout rate. Paired mean differences use 2,000 stratified bootstrap replicates [15] that resample matrices within the 16 equal-sized strata, and the 95% percentile intervals describe simulation uncertainty only. No weights are fitted to these results. The analysis is exploratory and not externally preregistered; the local protocol records two harness corrections (enforcing the unsure contract and reusing the production confidence summarizer).

C. Aggregate Results and Non-Superiority

Table I reports all six configurations over 24,576 runs. The deployed selector asks 10.07 questions on average, but that is not efficiency: in 47.02% of runs it ends the session while an eligible alternative is hidden by preference filtering. In our data these failures occur only in the six-item banks (94.04% of those runs, against none with three items per skill). The eligibility-first ordering removes the failure, asking 12.00 questions on average and reaching 33.81% assessed-skill coverage against 26.55% for the deployed policy. That

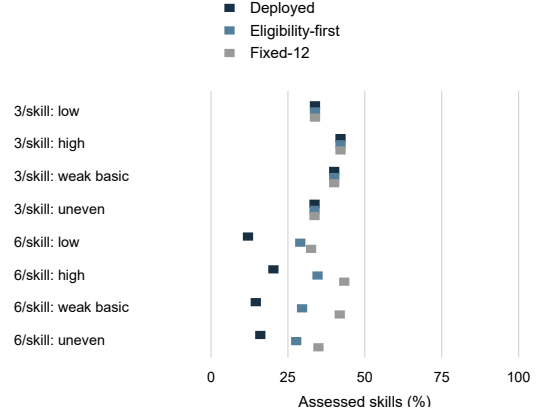


Fig. 3. Synthetic coverage by bank depth and response profile; the two noise levels are equally pooled. Each profile-depth group contains 512 matrices. No human accuracy is represented.

paired change of 7.26 percentage points has a synthetic 95% bootstrap interval of [6.87, 7.65].

Yet fixed stratified selection reaches 37.77% at twelve questions, so eligibility-first differs from it by -3.96 points, and the study does not support an adaptive coverage advantage. Fixing an availability defect and beating a reasonable baseline are different requirements. At question counts matched to the deployed policy, deployed minus stratified coverage is -3.91 points $[-4.33, -3.50]$. This holds burden closer to constant, but the matched budget is itself set by a failure-prone adaptive run, so it is not a randomized comparison of fixed budgets and says nothing about learning gains.

D. Sensitivity, Uncertainty, and Invariants

Removing self-report changes mean coverage by 0.00 percentage points [0.00, 0.00], and removing the graph-aware components changes it by 1.35 points [0.86, 1.88]. These are joint effects of the configured metadata, role filters, caps, and stopping policy, and a small or zero difference does not show that self-report or prerequisites are unimportant in real curricula. Because the cold-start design holds missing-mastery uncertainty at its maximum and gives every item identical priors, the information proxy is constant and the ablations cannot rank the coefficients; Fig. 3 and the per-stratum results keep the profile and depth differences visible.

All configurations observe all four skills, yet considerable skill-level uncertainty remains. No configuration emits a positive READY or MASTERED declaration, because diagnostic-only confidence cannot exceed 0.45 while the lowest positive threshold is 0.50. That zero is structural, not a measured false-positive rate; a learner study must measure whether knowledgeable learners can progress without excessive follow-up burden. A finite check enumerates 136 valid cases over one to four observations, four representative metadata schedules, binary correctness patterns, and all-unsure responses; none violates the confidence ceiling, the one-response PROBED rule, or the bounds of the uncertainty band, and none emits a posi-

tive cold-start classification. Planner traversal visits 100, 1,000, and 5,000 artificial skills in 25, 250, and 1,250 sessions within the four-skill scope, with selected scopes marked resolved by the harness; this shows first-pass scheduling reachability only, and we make no claim about latency, throughput, or course completion.

VII. DISCUSSION AND VALIDATION BOUNDARY

A. Interpretation

A shorter assessment can signal selection failure rather than sufficient evidence. The lesson is to separate physical bank exhaustion from candidate exhaustion caused by the policy: preference heuristics should operate inside an eligibility-preserving set so that a preferred but unusable skill cannot hide valid alternatives. This follows from the failure we demonstrated and is not a claim of universal optimality; availability, coverage, burden, estimation validity, and educational effectiveness are separate dimensions.

B. Threats to Validity

The simulator omits learning, fatigue, response time, correlated misconceptions, wording effects, and context-dependent transfer. The design space is narrow (four skills, two bank depths, fixed item priors, propensity-aligned self-report), the replay bypasses persistence, the interface, network failures, and concurrent learners, and finite invariant checks cannot replace integration tests, so the correction needs application-level verification before deployment. ASSESSED is an internal coverage state that includes confidently identified gaps, not an accuracy label. The confidence ceiling structurally rules out positive cold-start classifications and may trigger unnecessary reassessment. The uncertainty band is not a calibrated interval, and the bootstrap intervals describe this generator, not a learner population.

C. Independent Learner Evaluation

A follow-up study should recruit consenting learners and use expert-reviewed constructs and independent reference assessments that share no items with the system. It should prespecify an endpoint balancing incorrect readiness, unresolved placement, and question and time burden; compare corrected adaptive and stratified selection at comparable budgets; separate cold-start from returning learners; and estimate uncertainty at the learner level, with sample size from an endpoint-specific power analysis rather than the item-monitor gate of 20 responses. Retention and transfer claims need delayed checks and replication across subjects.

No participants were recruited and no private learner records were analyzed. Ethics approval, consent, and expert validation are not represented as completed. The local package provides code, source hashes, and synthetic outputs, and has no archival DOI. Human-effectiveness claims require a separate approved study.

VIII. CONCLUSION

LearnPath’s architecture makes the limits of its evidence inspectable, but that does not establish diagnostic validity. On RQ1, replay exposes a real preference-filtering defect, and eligibility-first ordering eliminates the premature null we observed. On RQ2, coverage improves over the deployed selector, yet fixed stratification stays higher at the same twelve-question budget, and the matched comparison also disfavors deployed adaptive coverage. On RQ3, the bank and profile results and the ablations describe behavior specific to this configuration, not the general value of any feature. Together, the architecture, the availability invariant with its correction, and a reproducible evaluation that keeps its negative results support an engineering fix, not adaptive superiority or diagnostic accuracy. Application-level integration and independent human validation remain future work.

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